

2.0 INTRODUCTION

The two most important data sets describing a loudspeaker driver are its frequency response (both magnitude and phase) and its *Thiele/Small (T/S) parameters*. Driver frequency response measurements will be discussed in Chapter 4. The T/S parameters of a driver are needed to correctly design an enclosure.

In this chapter I will give a very brief historical development of low-frequency direct-radiator loudspeaker system design, which highlights the introduction of the T/S approach, define the T/S parameters, give a brief development of their meaning from physical principles, and present techniques for measuring them. Because the driver impedance curve plays a major role in determining T/S parameters, its nature will also be discussed from first principles. Use of the T/S parameters in low-frequency direct-radiator loudspeaker system design is well covered in the *Loudspeaker Design Cookbook (LDC)*.¹ The reader is assumed to be familiar with this work.

2.1 A LITTLE HISTORY

From the invention of the electrodynamic loudspeaker by Kellogg and Rice² (circa 1920) until the 1970s, direct-radiator loudspeaker system design was as much an art as a science. The critical relationships between driver parameters, enclosure design, and loudspeaker system low-frequency response were poorly understood. A direct-radiator loudspeaker is a complex device first involving a transformation of electrical energy to mechanical energy followed second by a conversion of mechanical energy to acoustic energy. Although equations describing this multistep process were developed in the 1930s (e.g., McLachan³), these equations were complex and extremely difficult to solve. General solutions suitable for design purposes were not available.

In the 1940s, Olson⁴ and others introduced the concept of the analogous electric circuit to the low-frequency analysis of acoustic systems. In this approach acoustical elements are modeled as combinations of resistors, inductors, and capacitors. This is possible because under certain reasonable assumptions the *low-frequency* behavior of the acoustical elements obeys the same equations as their corresponding electrical elements. This meant that a complex system of acoustic elements could be modeled as an electric circuit and all the power of a highly developed electric circuit theory could be brought to bear on the analysis of loudspeaker system performance. Complete electric circuit models for closed-box and vented loudspeaker systems were given by Olson⁵ in 1943.

Unfortunately, as promising as the analogous circuit approach was, a big problem still remained. Some of the acoustical elements are frequency dependent. This means that the value of the analo-

gous electrical element must change with frequency. This greatly complicated the analytic solution of the circuit equations, and numerical solutions were out of the question without modern computers. Much of the frequency dependence could be ignored, but it was widely believed that one critical parameter, *radiation impedance*, could not. Radiation impedance, and in particular, the radiation resistance of a driver cone, is the parameter that relates cone velocity and acoustic output power. Ignore radiation resistance and you have no sound!

Using Olson's analogous circuit for the bass-reflex system, Beranek,⁶ in 1954, was able to solve the bass-reflex system response equations at three critical points where the radiation resistance could be calculated, but a complete response curve was still lacking. A major breakthrough in loudspeaker system analysis came in 1959.

Recognizing that direct-radiator loudspeaker systems are very inefficient (typical efficiencies are 1% or less), Novak⁷ reasoned that the effect of radiation resistance loading on the system could be ignored in the calculation of cone velocity. Once the velocity is known, acoustic output could then be calculated using the frequency-dependent radiation resistance. This led to a relatively simple mathematical solution for frequency response. Novak then analyzed the response of closed-box and vented systems and presented an analysis of the relative performance of the two.

An engineering discipline is not truly mature until it moves from analysis to synthesis. Given a set of system performance specifications, analytical tools allow a designer to converge on the correct design through a series of trials. After each trial, driver and/or enclosure parameters are changed and the analysis repeated until the desired performance is attained. Starting with a set of system performance requirements, synthesis tools lead directly to the required design. The birth of the synthesis approach to direct-radiator loudspeaker design is generally attributed to A.N. Thiele.

In 1961 he published a landmark paper entitled "Loudspeakers in Vented Boxes."⁸ Building upon Novak's work, Thiele showed how to select driver parameters, enclosure volume, and tuning so as to match or *align* a vented-box system-frequency response with that of a number of well-known electrical filter responses. Since frequency, phase, and transient response of the electrical filter are known, it followed that a loudspeaker aligned to a particular filter would have the same response. Thiele provided a table of nine discrete alignments for the vented enclosure.

He also presented some relatively simple procedures for measuring the primary driver parameters (now called Thiele/Small parameters) needed for the design process, namely resonant frequency, f_s , the Q , and equivalent volume, V_{AS} , using only driver electrical impedance data. With Thiele's paper, the loudspeaker system designer now had a systematic procedure for specifying driver parameters and enclosure design (volume and tuning) for vented systems to achieve a desired response.

At this point we might think that the analysis and synthesis of direct-radiator loudspeaker systems was

well-in-hand, but publications in professional society technical journals often trickle down slowly into the design community. Thiele's paper was published in an Australian technical journal and it was several more years before his results were widely known.

Over the period 1968–1972, J.E. Benson published a series of papers building on the works of Novak, Thiele, and many others, entitled, "Theory and Design of Loudspeaker Enclosures."⁹ This three-part series of some 240 pages provided a very general theory for all closed, vented, and aperiodically damped enclosures. Unfortunately, Benson's papers were also published in an Australian journal and languished for many years until they were referenced in Small's papers, discussed next.

The event that brought low-frequency, direct-radiation loudspeaker system synthesis to the attention of the general design community was the publication of R.H. Small's papers,^{10–12} in the *Journal of the Audio Engineering Society (JAES)*. Small, Benson's Ph.D. student, published a series of nine articles on the analysis and design of closed-box, vented, and passive-radiator systems. These papers expanded upon Thiele's work, providing detailed proofs of many of his results and a wide range of design examples. Small introduced the *continuous alignment graph* which replaced Thiele's limited discrete alignment table. With this plot a designer could find the proper enclosure volume (and tuning for vented systems) for a continuous range of driver parameters. Because the JAES had international distribution, Small's results were widely disseminated and by the mid-'70s, many designers were applying his design methodology with great success.

2.2 THE THIELE/SMALL PARAMETERS

Almost all of the useful driver parameters had been defined by other researchers and authors before Thiele and Small. However, they brought

these parameters and a few new ones together in a complete design approach and showed how they could be easily determined from impedance data alone. The parameters are now known collectively as the Thiele/Small parameters.

Figure 2.1 presents a cutaway view of a generic low-frequency driver, i.e., a woofer. This figure is taken from Reference 1 and should be familiar to the reader. With reference to this diagram, a partial list of T/S parameters is given below. Their meaning and physical significance will be discussed a bit later in the chapter.

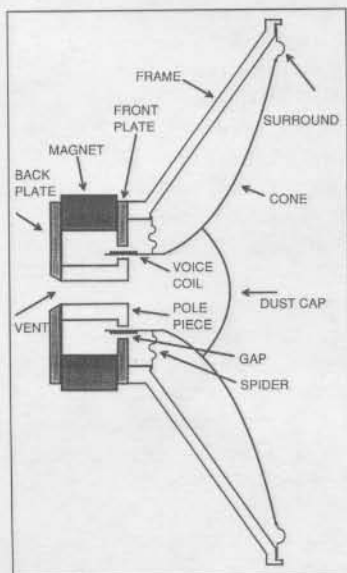


FIGURE 2.1: Cutaway view of a typical low-frequency driver.

The T/S parameters of primary interest, listed in alphabetical order, are:

B	Magnetic flux density in the driver gap
Bl	Motor strength—the product of B times l
C_{AS}	Acoustic compliance of the driver suspension
C_{MS}	Mechanical compliance of the driver suspension
f_s	Resonant frequency of driver including air load
f_{SA}	Free-air resonant frequency of driver
l	Length of voice coil wire in the magnetic field gap
L_{VC}	Voice coil inductance
M_{AS}	Acoustic mass of driver cone assembly including reactive air load
M_{MD}	Mechanical mass of the driver cone assembly, excluding air load
M_{MS}	Mechanical mass of diaphragm cone assembly including air load
Q_{ES}	Q of driver at f_s considering electrical resistance only
Q_{MS}	Q of driver at f_s considering mechanical losses only
Q_{TS}	Q of driver at f_s considering all driver losses
R_{AS}	Acoustic resistance of driver suspension losses
R_E	DC electrical resistance of voice coil
R_{ES}	Reflected electrical resistance due to driver suspension losses
R_{MS}	Mechanical resistance of driver suspension losses
S_D	Effective surface area of the driver cone
S_0	Reference voltage sensitivity of the driver
S_P	Reference power sensitivity of the driver
V_{AS}	Volume of air having the same acoustic compliance as driver suspension
η_0	Reference efficiency of driver

In the above list, the letter A used in the first position of the subscript refers to acoustic parameters. Electrical and mechanical parameters are denoted by E and M, respectively.

2.3 THE LOUDSPEAKER DRIVER IMPEDANCE CURVE

As a complex electro-mechano-acoustic transducer, a loudspeaker driver is characterized by a mix of electrical, mechanical, and acoustic parameters, many of which were listed in the previous section. The beauty of the T/S theory is that all driver parameters needed for loudspeaker system design can be determined from the *driver impedance curve* measured under just two different conditions. That is, all parameters of interest can be determined from the electrical or input side of the driver. Really rather amazing when you stop to think about it.

Because of its importance, let's spend a little time understanding the driver impedance curve and where it comes from. Figure 2.2 shows a typical low-frequency driver impedance plot. Both impedance magnitude and phase are shown. At very low frequencies the impedance magnitude is equal to the voice coil resistance, R_E , and its phase is close to 0°. The impedance rises with increasing frequency and the phase angle goes positive. In this region,

below driver mechanical resonance, the driver electromotive, or motional, impedance looks inductive. The impedance curve then peaks at a value of $R_E + R_{ES}$ at the driver mechanical resonant frequency, f_s , where the phase goes through zero again.

Above resonance, the plot falls and the phase angle goes negative. In this region, the driver impedance looks capacitive. Next, the curve reaches a minimum, where the voice coil inductance, L_{VC} , forms a series resonance with the capacitive electromotive impedance of the driver. Finally, the impedance curve rises again due to voice-coil inductance and other frequency-dependent effects discussed a bit later in this chapter. These effects dominate driver impedance from this point on.

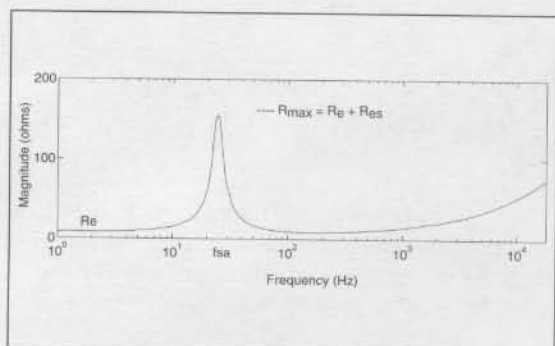


FIGURE 2.2a: Impedance magnitude of a typical 8" driver.

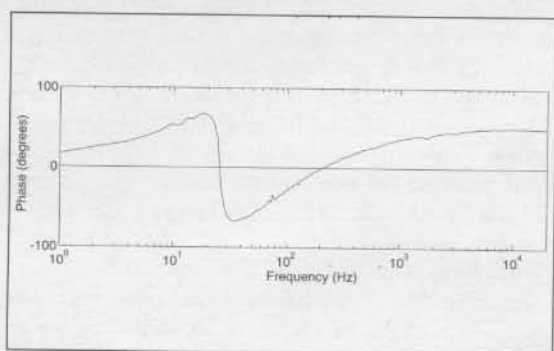


FIGURE 2.2b: Impedance phase of a typical 8" driver.

2.4 A SIMPLE DRIVER MODEL

Why does the impedance curve look like the plot shown in Fig. 2.2? Let's build up a simple model for this woofer that we can use to explain the shape of the impedance curve.

2.4.1 MODELING THE MECHANICAL SIDE

Figure 2.3 depicts a weight of mass, M , suspended from a fixed surface; for example, a ceiling, on a spring of stiffness, K . The mass is specified in kilograms. K has the units of force over distance. For instance, if it takes a force of one newton (1N) to stretch the spring from its resting length by one meter (1m), then K has the value of one newton/meter (1N/m).

Figure 2.3 also shows a dashpot damping element, D , which has units of force over velocity or newtons per meter per second. That is, the damping force restraining motion of the mass, M , is proportional to the velocity of the mass. (In this book we will often use the Systeme Internationale

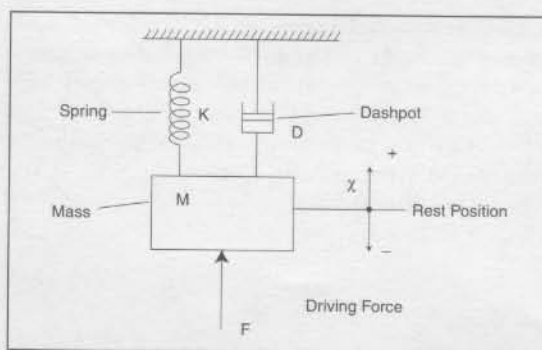


FIGURE 2.3: Mass/spring/dashpot model of driver mechanical properties.

(SI) set of units. The basic units of length, mass, and time in this system are the meter, kilogram, and second. The unit of weight or force is the Newton. Appendix A gives the conversion factors between Imperial and SI units.)

If the mass is pulled down by a fixed amount, potential energy is stored in the stretched spring. If the mass is now released it will bob up and down. In this idealized model the mass will bob up and down forever if D is zero; that is, if there are no mechanical losses in the system. The distance M moves from its rest or initial position—denoted by $x(t)$ in the figure—will vary sinusoidally in time. This is easily seen by referring to Fig. 2.4.

If the attachment point of the spring at the ceiling is moved at a constant velocity and a light is shown on the moving mass, the mass's shadow will trace out a sine wave on the wall behind it, much like the trace on an oscilloscope. The amplitude of this harmonic motion will change in proportion to the size of the initial displacement, but the frequency or period, T , will not. The frequency will be the same regardless of how far the mass is initially displaced as long as the spring is not stretched beyond its elastic limit.

The mass/spring combination forms a harmonic oscillator. The frequency of oscillation is given by the equation:

$$f_s = \frac{1}{2\pi} \sqrt{\frac{K}{M}} \quad [2.1]$$

where:

f_s = resonant frequency in Hz

K = spring stiffness in newtons/meter

M = mass in kilograms

Equation 2.1 tells us that a stiffer spring produces a

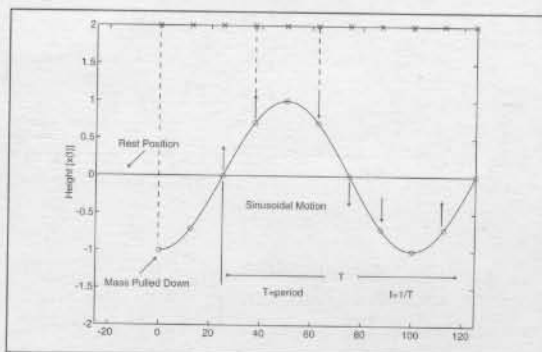


FIGURE 2.4: Periodic motion of spring/mass system.

higher resonant frequency, while a larger mass results in a lower f_s . This jibes well with intuition.

The concept of spring stiffness is intuitively appealing. In the loudspeaker world, however, it is customary to talk about spring or suspension *compliance*. Mathematically, compliance, C , is the inverse of stiffness.

$$C = \frac{1}{K} \quad [2.2]$$

Thus a very stiff spring has a low compliance and vice versa. Compliance has the units of distance over force or meters/newton. In terms of compliance, Equation 2.1 becomes:

$$f_s = \frac{1}{2\pi} \sqrt{\frac{1}{CM}} \quad [2.3]$$

M_{MD} is the total moving mass of the cone assembly. It includes the mass of the cone, voice-coil former and wire, the dust cap, and also a portion of the surround and spider which move with the cone assembly. There is an additional effective mass due to loading on the cone as it pushes against the surrounding air. This "air-load," referred to as mass reactance loading, will be discussed in more detail in Chapter 3. For now we simply note that the additional effective mass due to air load on a driver suspended in free air with no additional baffling is calculated with the following formula:

$$M_{M1} = 2.67\rho a^3 = 0.394D^3 = 0.566S_D^{1.5}$$

where:

a = driver cone radius

D = driver cone diameter

ρ = density of air (1.18kg/m³ at 20°C sea-level)

Then the total moving mass including air load becomes:

$$M_{MS} = M_{MD} + M_{M1}$$

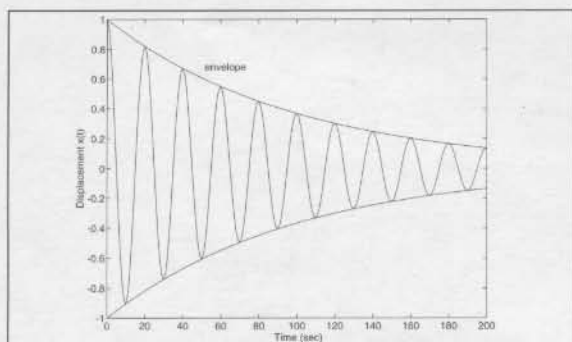


FIGURE 2.5: Decaying oscillation and its envelope.

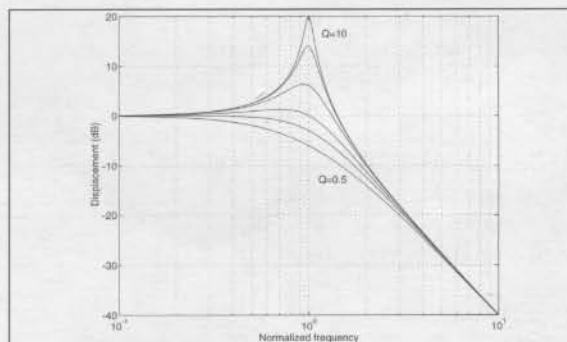


FIGURE 2.6a: Normalized displacement versus mechanical Q.

C_{MS} is the total compliance of the cone assembly and includes the compliance of the spider and surround. If we replace C and M in Fig. 2.2 with C_{MS} and M_{MS} , we have a simple model for the mechanical portion of a driver, and the equa-

tion for the free-air resonance becomes:

$$f_{SA} = \frac{1}{2\pi} \sqrt{\frac{1}{C_{MS}M_{MS}}} \quad [2.4]$$

The damping, D , provides the mechanism for dissipating energy in the harmonic oscillator. If D is not zero, the oscillation excited by the initial displacement will decay to zero exponentially in time (Fig. 2.5). In addition to the decaying oscillation, Fig. 2.5 shows two exponentially decaying curves, one above and one below the actual oscillation. These curves define an envelope that bounds the peak magnitude of the oscillation over time. The rate of decay of the envelope is described by the nondimensional parameter, Q_{MS} .

$$Q_{MS} = 2\pi f_s \left(\frac{M_{MS}}{C_{MS}} \right) = \left(\frac{1}{D} \right) \sqrt{\frac{M_{MS}}{C_{MS}}} \quad [2.5]$$

Q_{MS} is inversely proportional to D . Thus, a large value of damping is equivalent to a low value of Q_{MS} and vice versa. The rate of decay of the oscillation in dB per cycle is:

$$DR = -54.6/Q_{MS} \text{ dB/cycle; } Q_{MS} > 0.5 \quad [2.6]$$

Equation 2.6 is valid for values of Q_{MS} greater than 0.5. Below this value the system is overdamped and an initial displacement decays uniformly to zero without oscillation. From Equation 2.6 it is clear that a system with a high Q will decay slowly in time, while a system with low Q will die rapidly. Q_{MS} ranges from 3-10 for commonly available drivers. Much more will be said about Q shortly.

2.4.2 THE FORCED SYSTEM

So far we have discussed what happens over time when the system of Fig. 2.3 is disturbed by an initial displacement. What happens when the system is driven continuously with a sinusoidal force of constant amplitude and varying frequency? The displacement frequency response of the driven mass is shown in Fig. 2.6a for values of Q_{MS} of 0.5, 0.7, 1, 2.5, and 10. At frequencies much below resonance, the displacement response is constant. In this frequency region, the forcing motion is resisted only by the spring stiffness. This is referred to as the stiffness- or compliance-controlled region.

As the exciting frequency is increased, motion of the mass increases until it reaches a maximum at or slightly above f_s , depending on the value of Q_{MS} . From the graph it is clear that the peak of the response is controlled by the value of Q_{MS} , smaller

values implying greater damping and less peaking. The region around f_s is said to be damping controlled. Somewhat above f_s response falls off, reaching a constant rate of -12dB/octave . Here inertial forces dominate as it becomes increasingly harder to move that mass with increasing frequency. This is called the *mass-controlled region*.

Figures 2.6b and 2.6c give more insight into the frequency response of the system. Figure 2.6b depicts the velocity response of the driven mass. This curve has a bandpass characteristic, rising 6dB per octave at low frequencies, reaching a maximum at f_s and falling at 6dB per octave above f_s . This explains why the damping force, which is proportional to velocity, dominates in the frequency region around f_s , where velocity is highest. Unlike position response, velocity response is *always* maximum at f_s where velocity is highest. This fact will be used later to determine f_{SA} from the impedance curve.

Acceleration response is shown in Fig. 2.6c. Acceleration rises at 12dB/octave below f_s . Above f_s , however, acceleration flattens out and becomes constant. In order to keep the mass displacement response constant, acceleration must increase as the square of frequency.

Conversely, if acceleration is constant, displacement must fall as one over the square of frequency or 12dB/octave as was shown in Fig. 2.6a. Notice that the acceleration response plot is similar to the low-frequency response of a closed-box speaker system (see Fig. 1.1 of Reference 1). This is no coincidence because the pressure response of a loudspeaker is proportional to driver cone acceleration.

2.4.3 WHAT IS THIS THING CALLED Q?

In the early days of radio, parallel resonant LC tank circuits were used extensively to improve the frequency selectivity of radio-frequency receivers. (It is unfortunate that the symbol C is used for both mechanical compliance and electrical capacitance. The meaning of C should be clear by the context.) These tank circuits have a bandpass frequency response just like the velocity response of our spring/mass system. The narrower the peak in response, the more selective the circuit is.

The selectivity of the LC tank circuit is characterized by a nondimensional factor, the *quality factor*, or Q , of the circuit. Q s of 50, 100, or more were common for these circuits. A higher Q meant less damping or electrical loss in the circuit, and therefore, higher selectivity. Q s were limited by such factors as wire resistance in the coils and the dissipation factor of the capacitors of the day.

Q is often referred to as a measure of the energy lost by a system, but this is only half the story. Formally, Q is defined as a ratio:

$$Q = \frac{\text{energy stored per cycle}}{\text{energy dissipated per cycle}},$$

where "per cycle" refers to one cycle of the sine wave driving force, F . The driving force supplies the input energy to the system. Q is thus a measure of how much of the input energy is retained by the system versus how much is dissipated ultimately as

heat. If more energy is retained than lost, Q will be greater than one.

Clearly, this definition of Q can be applied to mechanical or electrical systems. For the mechanical system of Fig. 2.2, energy is stored alternately as potential energy in the compression or elongation of the spring, or as kinetic energy in the velocity of the mass. Energy is dissipated through the action of damping. In a driver, mechanical losses are produced by the flexing of the spider and surround. Later in this chapter, an electrical Q will be defined, which accounts for losses on the electrical side of the driver.

We now have a complete model for the mechanical side of a driver, so let's turn our attention to the electrical side.

2.4.4 MODELING THE ELECTRICAL SIDE

The voice coil together with the magnetic field in the voice coil gap form a linear motor. Figure 2.7 is an expanded view of the voice coil and annular voice coil gap geometry. The magnetic flux lines are radial in the gap and perpendicular to the voice-coil windings. It follows that they are also perpendicular to any current flowing in the wire. When a current, i , is passed through the voice coil, a force, F , is produced and that drives the cone. The force is proportional to the magnetic flux density, B , the length of wire in the magnetic field, l , and the current, i . The appropriate equation is:

$$F = Bli \quad [2.7]$$

The direction of the force is determined using the procedure illustrated in Fig. 2.8a. Draw an arrow in the direction of the magnetic flux lines. Draw a second arrow in the direction of the current flow. Rotate i into B through the smaller angle

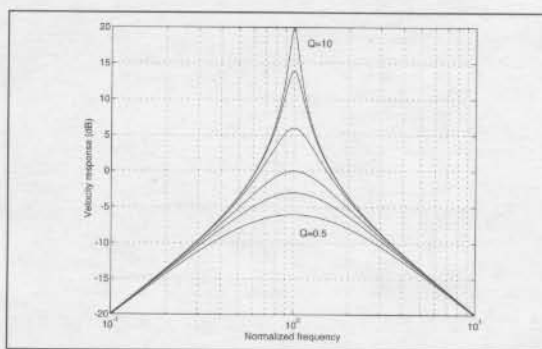


FIGURE 2.6b: Velocity response versus mechanical Q .

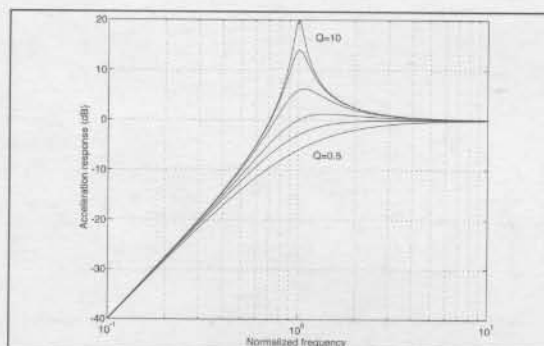


FIGURE 2.6c: Acceleration response versus mechanical Q .

between them. The force will be in the direction that a right-hand screw would move with that same rotation. For example, if i must rotate clockwise to meet B , a right-handed screw would advance into the plane of the paper. Known as the *right-hand rule*, it shows that the resulting force is always perpendicular to both the current direction and the flux lines; that is, it is always perpendicular to the plane formed by the i and B arrows.

For those of you familiar with vector algebra, the force F is computed as the vector cross product of i and B , where i is taken in the direction of the conductor, l . For a driver, the force acts along the voice-coil axis and reverses with reversal in the direction of the current. The linear motor pushes the voice coil and thereby the cone in and out relative to the driver frame.

At first glance it might appear that the voice-coil current is limited only by voice-coil impedance and the voltage applied to the coil terminals, but something else is going on here. The voice coil moving

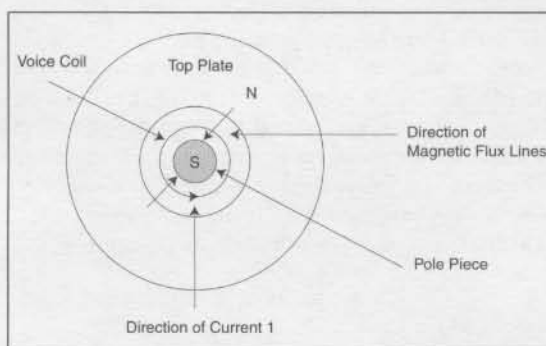


FIGURE 2.7: Voice-coil magnetic gap geometry. Force is into the paper for the direction of current flow and flux lines.

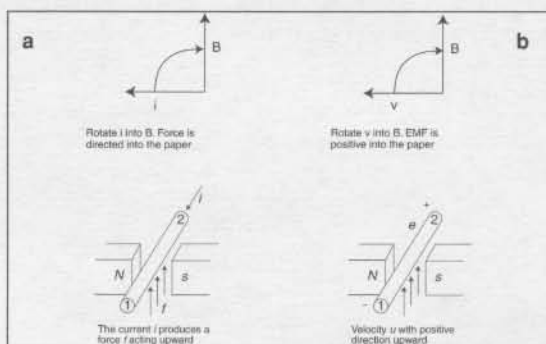


FIGURE 2.8: Illustrates the direction of force and back EMF.

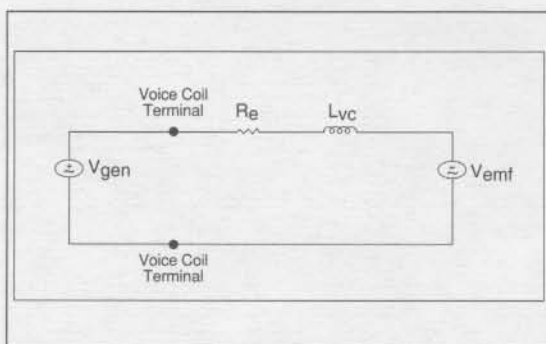


FIGURE 2.9: Equivalent circuit seen at voice-coil terminals.

through the magnetic field acts like a generator. According to one form of Faraday's Law, when a conductor of length, l , moves at right angles to a magnetic field, B , at a velocity, v , an electromotive force (EMF) or voltage is generated across the conductor of magnitude:

$$EMF = Blv \quad [2.8]$$

The polarity of this voltage is shown in Fig. 2.8b. It is determined by drawing the arrows or vectors v and B and then rotating v into B through the smaller angle between them. The polarity of the back EMF again corresponds to the direction in which a right-hand screw would move. Notice that the polarity of the generated voltage causes a current to flow which opposes the current flow induced by the applied voltage, V_{gen} . For this reason it is referred to as a *counter*, or *back EMF*.

2.4.5 THE DRIVER IMPEDANCE CURVE

We are now in a position to explain the driver impedance curve. An equivalent circuit looking into the voice-coil terminals is shown in Fig. 2.9. Assume an AC sine-wave voltage of *constant amplitude*, V_{gen} , is applied to the voice-coil terminals (ignore voice-coil inductance for now). In response to the applied voltage, a current will flow in the voice coil, in turn generating a sinusoidally varying force, causing the cone to move back and forth. From Ohm's Law, the current, i_{VC} , at very low frequencies will equal:

$$i_{VC} = \frac{V_{gen}}{R_E}$$

As the sine-wave frequency is increased, however, cone velocity will increase according to Fig. 2.6b, reaching its maximum value at f_{SA} . Likewise, the back EMF producing the opposing current flow will increase and thus the net current in the voice coil will decrease to a minimum at f_{SA} . As frequency is further increased, velocity and back EMF will fall off, ultimately allowing the current to return to its low-frequency value (remember L_{VC} is ignored for the moment). A typical current versus frequency profile is illustrated in Fig. 2.10a.

For a different look at the driver impedance function, now assume that the applied voltage amplitude is varied with frequency so as to main-

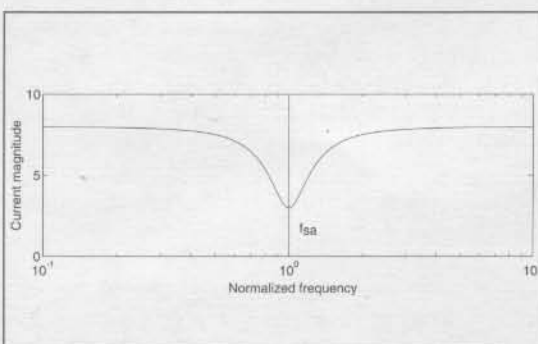


FIGURE 2.10a: Voice-coil current with constant-voltage sweep.

tain a constant current in the voice coil. Again from Ohm's Law, the current, i_{VC} , is:

$$i_{VC} = \frac{V_{gen} - V_{emf}}{R_E} \quad [2.9]$$

As frequency is increased, the back EMF will again increase, and therefore, applied voltage (V_{gen}) must increase to keep the current constant. The applied voltage will reach a maximum at f_{SA} and fall off as frequency is increased beyond f_{SA} . A plot of the applied voltage amplitude versus frequency is shown in Fig. 2.10b.

Since voice-coil impedance is defined as V_{gen}/i_{VC} , it should be clear from the previous discussion and Fig. 2.10 that voice-coil impedance reaches a maximum at f_{SA} , assuming L_{VC} is zero. Back EMF is the mechanism by which the mechanical and acoustical properties of a driver are reflected back into the electrical impedance seen at the voice-coil terminals. Thiele and Small showed us how to extract the T/S parameters from the impedance data.

The impedance rise in the vicinity of f_{SA} is identical to that of a parallel resonant RLC circuit. It follows that an analogous all-electrical circuit model for voice-coil impedance can be constructed. Figure 2.11 gives the complete electrical model for driver impedance considering all electrical and reflected mechanical and acoustic effects. In Fig. 2.11 the electrical capacitance, C_{MES} , represents the total moving mass of the driver—including air load—reflected back to the electrical side. (See Chapter 3 for a discussion of the difference between M_{MS} and M_{MD} .) C_{MES} is related to the moving mass by:

$$C_{MES} = \frac{M_{MS}}{(Bl)^2} \quad [2.10]$$

L_{CES} represents the driver's mechanical compliance reflected back to the voice-coil terminals and is given by:

$$L_{CES} = C_{MS}(Bl)^2 \quad [2.11]$$

Finally, R_{ES} represents the mechanical losses reflected back to the electrical side. It is computed by:

$$R_{ES} = \frac{(Bl)^2}{R_{MS}} \quad [2.12]$$

Notice in Equations 2.10–2.12 that all conversions of mechanical parameters to the electrical side involve the factor $(Bl)^2$. The first " Bl " comes from the " Bl " effect while the second one comes from the " Blv " term. Notice also that the complete circuit includes frequency-dependent resistance and inductance indicated in the drawing as R_{fd} and L_{fd} . That is, both the resistive and inductive portions of the equivalent circuit vary with frequency.

Audio currents passing through the voice coil

produce an AC magnetic field, which in turn induces circulating currents in the pole piece, called *eddy currents*. The eddy currents generate heat that increases with increasing frequency. The net result is an increasing resistance and a decreasing inductance as the applied frequency rises. This effect will be discussed in more detail later in this chapter.

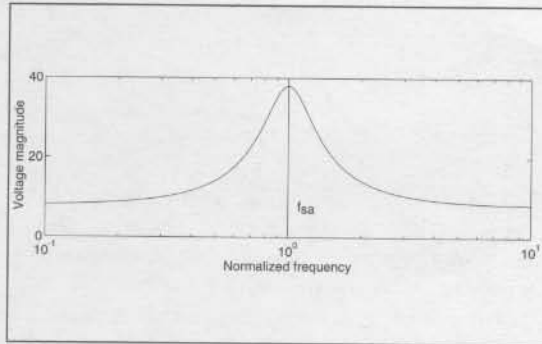


FIGURE 2.10b: Voice-coil voltage with constant current sweep.

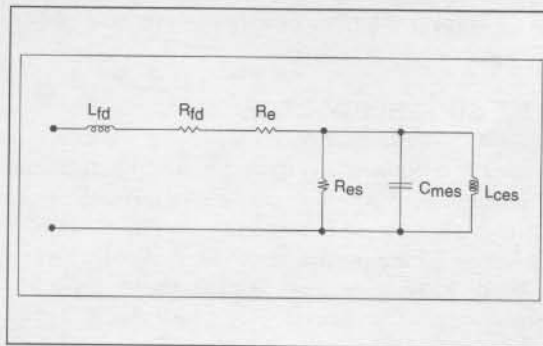


FIGURE 2.11: Equivalent electric circuit for driver impedance.

2.5 A BIT MORE ON T/S PARAMETERS

Before getting to actual parameter measurement techniques, two more parameters need to be defined and their significance explained. They are Q_{ES} and V_{AS} .

2.5.1 BACK EMF, ELECTRICAL DAMPING, AND Q_{ES}

Back EMF produces electrical damping which augments mechanical damping in a driver. In fact, for drivers commonly used in direct-radiator loudspeakers today, electrical damping is often the dominant source of damping, exceeding that from mechanical and acoustical losses.

To see how back EMF works, try the following experiment. Take a large, high-compliance woofer, preferably a 12" or 15" unit with a low f_s . Press down sharply on the cone near the dust cap. The cone will move easily. Now short the voice-coil terminals and again press down sharply. You will feel a much greater resistance to motion. Pressing down sharply on the cone imparts a high velocity to the voice coil, which in turn generates a back EMF.

This is easily observed by placing a voltmeter across the open voice-coil terminals. With the terminals shorted, however, the back EMF produces

a large current in the voice coil, and from Equation 2.7, a force which opposes the cone motion and thereby damps the cone. Loudspeakers are usually driven from low output impedance amplifiers in order to maximize the effect of electrical damping.

Thiele defined an *electrical Q*, Q_{ES} , to characterize this electrical damping. The equation for Q_{ES} is:

$$Q_{ES} = \left(\frac{R_E}{(Bl)^2} \right) \sqrt{\frac{M_{MS}}{C_{MS}}} \quad [2.13]$$

Physically, this equation tells us that Q_{ES} is inversely proportional to the square of the magnetic field strength so that a bigger magnet means more electrical damping. Q_{ES} increases with increasing R_E because a larger value of R_E reduces the current flow in the voice coil when its terminals are shorted and this, in turn, reduces the damping force. Q_{ES} increases with increasing cone mass simply because it is harder to damp a more massive cone. Finally, Q_{ES} decreases with increasing compliance because it takes less force to damp a more compliant suspension.

2.5.2 AIR SPRINGS, MECHANICAL COMPLIANCE, AND V_{AS}

Figure 2.12 shows two sealed boxes. Each box is equipped with a piston of area S_D forming an air spring. The second box has the smaller volume. If both pistons are pushed down the same distance, a greater force is needed to push down piston 2. Since we have defined a spring stiffness, K , as the

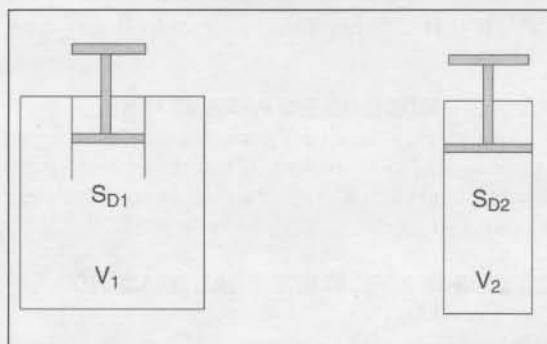


FIGURE 2.12: Effect of enclosed volume on air-spring compliance. V_1 is greater than V_2 , but S_{D1} equals S_{D2} . For this condition C_{M1} is greater than C_{M2} .

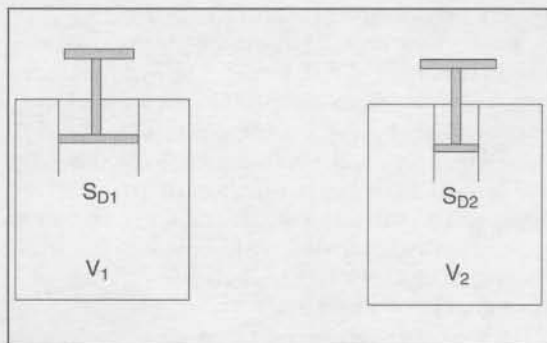


FIGURE 2.13: The effect of piston size on air-spring compliance.

ratio of force/distance, it is clear that air spring 2 is stiffer than air spring 1. Conversely, spring 1 is more compliant than spring 2. For a fixed piston area, the larger the enclosed volume, the higher its compliance becomes.

Figure 2.13 shows two boxes of equal volume, but now the piston in box 2 is smaller than that in box 1. If both pistons are again pushed down the same distance, a greater volume of air will be compressed by piston 1 relative to piston 2, and the force required to move piston 1 will be greater than that for piston 2. In this example, spring 2 is more compliant than spring 1. We see that air-spring compliance depends on both the enclosed volume and the piston area.

The air spring analogy can be used to specify a driver's compliance in terms of an equivalent volume of air. The appropriate parameter (V_{AS}) represents the enclosed volume of air which has the same compliance (or stiffness) as the suspension system of the driver when that air volume is compressed by a piston of the same area as the driver diaphragm. V_{AS} is a useful concept because it helps us to visualize the range of enclosure volumes in which a particular driver will work well. Specifically, V_{AS} is used to calculate the enclosure volume needed to obtain a particular response alignment using the α parameter described in Chapter 1 of *LDC*. The relationship between mechanical compliance (C_{MS}) and V_{AS} is given by the formula:

$$V_{AS} = \rho c^2 C_{MS} S_D^2 \quad [2.14]$$

where:

ρ = density of air (1.18 kg/m³ at 20°C sea-level)
 c = speed of sound (344.5ms at 20°C sea-level)

V_{AS} is an acoustic parameter. For a given mechanical compliance, the *acoustic volume*, V_{AS} , varies as the square of S_D . This explains why large-diameter high-compliance drivers need large enclosures. Also note that V_{AS} changes with changes in air temperature and atmospheric pressure because both ρ and c change with these variables.

2.6 DETERMINING BASIC T/S PARAMETERS FROM THE IMPEDANCE CURVE

As an absolute minimum, the parameters f_{SA} , Q_{TS} , and V_{AS} are required to design an enclosure for a driver. Measurement of these parameters will be discussed first. Thiele recommended that the test-driver be suspended in air as far from any sound-reflecting surfaces as possible.

The resulting measured parameters are called the "free-air" parameters. Thiele (and later Small) derived a series of equations for determining the T/S parameters from the simplified model of voice-coil impedance. We have already stated that radiation resistance can be ignored. Thiele also ignored the effect of voice-coil inductance, which is usually small in the frequency region around resonance. With these assumptions, a typical voice-coil impedance magnitude plot is depicted in Fig. 2.14.

Several features stand out in this figure. The

impedance magnitude plot reaches maximum value, $R_{\max}$, equal to $R_{ES} + R_E$ at f_{SA} and the phase is zero; that is, the impedance is purely resistive at f_{SA} . Notice that the impedance magnitude returns to R_E at both very low and very high frequencies. Finally, the impedance magnitude plot is symmetric about f_{SA} when plotted on a logarithmic frequency scale. This last observation implies that the curve has geometric symmetry about f_{SA} and for any two frequencies, f_1 and f_2 , symmetrically placed about f_{SA} on a logarithmic scale we have:

$$f_1 f_2 = f_{SA}^2 \quad [2.15]$$

This fact can be used later in testing to verify the accuracy of Thiele's simplifying assumptions.

Starting from Fig. 2.14, Thiele developed the following relationships:

First define:

$$r_0 = \frac{R_{\max}}{R_E} = \frac{R_{ES} + R_E}{R_E} \quad [2.16]$$

and

f_1, f_2 = frequencies below and above f_{SA} , respectively, where the impedance magnitude, R_x , equals $R_E \sqrt{r_0}$.

Then:

$$Q_{MS} = \frac{f_{SA} \sqrt{r_0}}{f_2 - f_1} \quad [2.17]$$

and

$$Q_{ES} = \frac{Q_{MS}}{r_0 - 1} \quad [2.18]$$

$$Q_{TS} = \frac{Q_{MS} Q_{ES}}{Q_{MS} Q_{ES}} = \frac{Q_{MS}}{r_0} \quad [2.19]$$

From these equations practical techniques for measuring f_{SA} and Q_{TS} can be developed. Referring to Fig. 2.14, we must measure R_E and three points on the impedance curve to determine f_{SA} and Q_{TS} . Three approaches will be discussed: the *voltage-divider procedure*, the *constant-voltage technique*, and the *constant-current technique*.

The voltage-divider approach uses the simplest instrumentation suite, but can be less accurate if not properly corrected. The constant-voltage procedure is more accurate, but requires a power amplifier. The constant-current technique uses an AC constant-current source (CCS). It is as accurate as the constant-voltage approach and somewhat quicker and easier to use, but requires the most sophisticated instrumentation.

use, and therefore, its parameters will shift. In order to account for this, it is important that all drivers be broken-in before testing. This can be accomplished by suspending the driver in free air and driving it with a power amplifier at a frequency in the 20–25Hz range. The drive level should be set for moderate cone travel. Be careful not to overdrive the speaker or physical damage may result. Break in the driver for at least one hour. Longer periods may be needed. Some manufacturers specify their driver parameters after eight and even 24 hours break-in.

The T/S parameters are “small signal” parameters. It is important in all the tests described below to keep drive levels as low as your instrumentation will allow while still providing reliable results. When suspended in air, a driver cone has little resistance to motion at low frequencies. Under this condition, relatively small drive levels can produce large cone/voice-coil excursions which push the driver suspension or motor into a nonlinear region. This is especially true when the driving frequency is at or near driver free-air resonance. T/S parameters measured under large excursions will differ from their “small signal” value.

2.6.1 TESTING PRELIMINARIES

The suspension on a driver will loosen up with